

Divide quantities in a given ratio



- 1 James and Emma scored goals in their netball game in the ratio 4 : 3.
 - a Find the fraction of the goals scored by James.
 - b If they scored a total of 56 goals, how many goals did James score?
 - c How many goals did Emma score?

- 2 Harris and Bella divided lollies between them in the ratio 5 : 4.
 - a Find the fraction of lollies that Bella gets.
 - b If there are 27 lollies, how many does Bella get?
 - c How many lollies does Harris get?

- 3 56 building blocks are shared between Mohamad and Isabelle in the ratio 2 : 5.
 - a Find the fraction of the blocks Mohamad receive.
 - b How many blocks does Mohamad receive?
 - c How many blocks does Isabelle receive?

- 4 Ben always buys melons and bananas in the ratio 7 : 3. If he buys 30 pieces of fruit in total, how many melons did he buy?

- 5 The ratio of adults to children on a train is 2 : 7. The train is carrying 360 passengers.
 - a How many adults are on the train?
 - b How many children are on the train?

- 6 James and Sophia share \$880 in the ratio 2 : 9.
 - a How much money does James get?
 - b How much money does Sophia get?

- 7 Avril is making a scarf that uses two materials. She wants to use the materials in the ratio of $\frac{11}{15}$ metres of red for every $\frac{3}{5}$ metres of green.
 - a Write the simplified ratio of red material to green material.
 - b If she needs 33 metres of red material, how many metres of green material will she need?

- 8 Neil is mixing paint to get a specific shade of orange. He wants a mix of red and yellow paint in the ratio of $3\frac{3}{4} : 6\frac{2}{3}$.
- Simplify the ratio.
 - How many litres of red paint will Neil need if he wants to make 75 litres of orange paint?
- 9 Kathleen and Luke invest money in a business in the ratio 3 : 10.
- What fraction did Kathleen contribute?
 - If Kathleen has invested \$2100, how much did Luke invest?
- 10 At a sporting event, the ratio of security personnel to spectators is 1 : 180. If 34 200 spectators attend the event, how many security personnel will be required?
- 11 Two kinds of pine trees, type A and type B, are planted in rows. In each row the ratio of type A to type B is 9 : 8 Altogether, 180 type A trees are planted.
- How many type B trees are planted?
- 12 Xavier and Ursula split the cost of the groceries in the ratio 0.5 : 1.5. Altogether the groceries cost \$140.
- Simplify the ratio.
 - How much did Xavier pay for the groceries?
- 13 A car and a truck are driving down the highway. Their relative speeds are in the ratio 3.1 : 2.9. If the car is travelling at 93 km/hr, how fast is the truck travelling?
- 14 In a zoo the ratio of elephants to lions is 7 : 4.
- Find the ratio of elephants to lions to the total number of elephants and lions.
 - Complete the following table of equivalent ratios:
- | Elephants | to | Lions | to | Total |
|-----------|----|-------|----|-------|
| 7 | : | 4 | : | |
| 14 | : | | : | |
| | : | | : | 110 |
- If there are 66 elephants and lions altogether, how many lions are there?

15 The ratio of red, green and black coloured  on a dart board is 12 : 7 : 31. The area of the dart board is 150 cm^2 .

a Find the number of ratio parts corresponding to the following:

- i** Red area **ii** Green area **iii** Black area **iv** Total area

b Find the size of a single ratio part.

c Find the area of the following:

- i** Red area **ii** Green area **iii** Black area

16 Vanessa competed in a marathon and found that her times for the swimming, cycling and running parts were in the ratio 3 : 1 : 2. If the quickest part took 10 minutes, how many minutes did she take to complete the whole marathon?

17 The ratio of students who play badminton, football and softball is 2 : 5 : 9.

a If there are 72 students who play softball, how many students play badminton?

b Find the total number of students who play one of these sports.

18 The perimeter of a rectangle is 42 cm and the ratio of its length to its width is 4 : 3.

a How many parts are in the ratio?

b Find the length of the rectangle.

c Find the width of the rectangle.

Increase and decrease in a given ratio (Extended)

19 Increase the following quantities in the ratio specified:

a \$400 in the ratio 3 : 1.

b \$780 in the ratio 5 : 2.

c 60 kg in the ratio 4 : 3.

d 800 g in the ratio 3 : 2.

e 1.5 L in the ratio 2 : 1.

f 1200 mL in the ratio 6 : 5.

g 120 cm in the ratio 10 : 2.

h 5.2 m in the ratio 100 : 5.

20 Decrease the following quantities in the ratio specified:

a \$500 in the ratio 1 : 5.

b \$820 in the ratio 3 : 4.

c 50 kg in the ratio 2 : 5.

d 150 g in the ratio 1 : 2.

e 2.4 L in the ratio 5 : 6.

f 900 mL in the ratio 4 : 5.

g 20 cm in the ratio 8 : 10.

h 2.4 m in the ratio 5 : 20.



21 A clothing store is having a sale and is decreasing the price of all their items in the ratio 3 : 4. Find the new sale price of the following items:

- a** A jumper originally priced at \$50.
- b** A dress originally priced at \$120.
- c** A pair of jeans originally priced at \$80.
- d** A skirt originally priced at \$75.

22 At the beginning of the year, an office supplies store increases the prices of its items in the ratio of 7 : 6. Find the new increased prices of the following items:

- a** A pencil case originally priced at \$12.
- b** A desk originally priced at \$180.
- c** A pair of jeans originally priced at \$54.
- d** A set of shelves originally priced at \$150.